GTS123 Introduction to Computers and Programming

Lab 9: Problem 9

Consider a cylinder which is defined by radius and height. Write a Python program that receives the radius (r) and height (h). The program uses a function called "myCylinder(r, h)" whose inputs are r and h to return the volume (v) and surface area (a) according to the following equations:

$$v = \pi r^2 h \qquad and \qquad a = 2\pi rh + 2\pi r^2$$

The output of the program should look similar to the following.

```
Enter the radius r of the cylinder: 1
Enter the height h of the cylinder: 3
The volume is 9.42 and the surface area is 25.13.
```

```
Enter the radius r of the cylinder: 3.3
Enter the height h of the cylinder: 4.5
The volume is 153.95 and the surface area is 161.73.
```

```
Enter the radius r of the cylinder: -2
Enter the height h of the cylinder: 4
Please enter positive values for both radius and height.
```

```
Enter the radius r of the cylinder: 5
Enter the height h of the cylinder: -1
Please enter positive values for both radius and height.
```

Note: To pass the test, your program must produce the exact output as shown above, including spaces and line breaks.